Aditya Chouhan

Software Developer Intern

Ujjain, M.P. | +91 7880209425 | adityachouhan2446@gmail.com | Portfolio | LinkedIn | GitHub

Professional Summary

Highly motivated Java Backend Developer and Computer Science engineering student (Expected Graduation: 2026) with hands-on expertise in engineering scalable microservices, distributed architectures, and high-throughput RESTful APIs. Proficient in modern Java ecosystems (Java 21, Spring Boot 3.3) and normalized relational database design (PostgreSQL, MySQL). Passionate about leveraging cloud-native technologies (Docker, Apache Kafka) and AI integrations (Spring AI) to solve complex system design challenges and build fault-tolerant web applications.

Education

Shri Vaishnav Vidyapeeth Vishwavidyalaya, Indore, MP **2022–2026**
Bachelor of Technology (B.Tech) in Computer Science & Engineering *CGPA: 6.98*

Ujjain Public School, Ujjain, MP **2021–2022**
Class XII (Senior Secondary) – Science (PCM), MP Board

Jolly Memorial Mission School, Ujjain, MP **2019–2020**
Class X (Secondary School), CBSE Board

Technical Skills

Programming Languages: Java (21/17), JavaScript (ES6+), C++, SQL, HTML5, CSS3.
Backend & Frameworks: Spring Boot 3.3, Spring Security, Spring AI, Hibernate/JPA, RESTful APIs, Microservices, JDBC, JSP.
Databases & Caching: PostgreSQL, MySQL 8.0 (3NF/ACID), MongoDB, Redis, Supabase.
DevOps & Tools: Docker, Apache Kafka, Git, GitHub, Maven, Gradle, Postman, JUnit 5, Vercel, Render.
Core Competencies: Data Structures & Algorithms, Object-Oriented Programming, Database Indexing, JWT & OAuth2, CI/CD Pipelines.

Projects

FairCart | *AI-Powered Distributed E-Commerce Platform* Live Demo | GitHub

Java 21, Spring Boot 3.3, Microservices, Apache Kafka, Redis, PostgreSQL, Docker

- Architected a decoupled microservices ecosystem (Auth, Catalog, Cart, Order, Inventory) utilizing Apache Kafka event streams and Dead-Letter Queues (DLQ) to ensure fault-tolerant, asynchronous order dispatching.
- Integrated Spring AI with Large Language Models (LLMs) to enable real-time dynamic deal negotiation, personalized product recommendations, and automated cart abandonment anomaly detection.
- Engineered a high-throughput distributed caching layer using Redis, reducing primary database read load by 55% and achieving sub-25ms response latency during simulated concurrent flash-sale traffic.
- Designed 3NF normalized relational schemas in PostgreSQL using Hibernate/JPA, implementing pessimistic locking to guarantee ACID compliance and eliminate race conditions during high-concurrency checkouts.
- Enforced stateless JWT and OAuth2 authentication with granular Role-Based Access Control (RBAC), automated unit testing via JUnit 5, and containerized the microservices environment using Docker.

FixOra | *On-Demand Local Service Booking Marketplace* Live Demo | GitHub

Java 21, Spring Boot 3.3, Spring Security, MySQL 8.0, Redis, Docker

- Built a production-grade service marketplace featuring dedicated, role-based dashboards (Customer, Service Pro, Admin) secured by stateless JWT authentication and BCrypt password encryption.
- Implemented location-aware service discovery by leveraging the Haversine formula across multiple city clusters, integrated with dynamic, collision-free slot reservation logic.
- Architected 17 normalized (3NF) database tables in MySQL 8.0 utilizing composite B-Tree indexing, foreign key cascading, and ACID-compliant stored procedures for optimized transaction auditing.
- Engineered an automated 5-stage job lifecycle pipeline (Booked to Paid) with real-time status tracking, and successfully containerized the application with Docker for seamless cloud deployment on Render.

Certifications

- Java Programming & Cloud Computing – NPTEL / IIT Kharagpur
- Relational Databases & MySQL – Oracle